

Introduction to Sets and Methods of Describing Sets

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A **set** is a well-defined collection of objects called **elements**.

Naming Sets

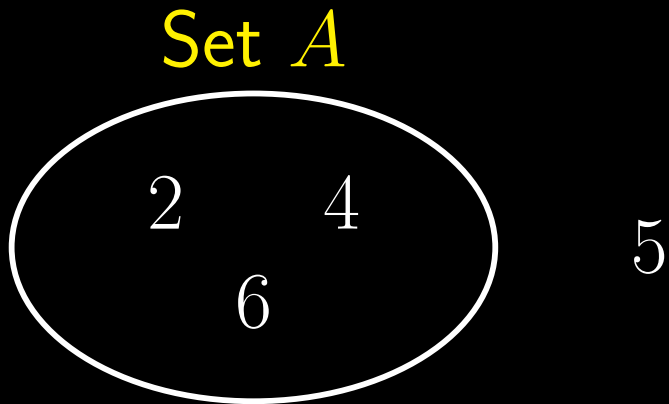
Sets are usually named using capital letters, such as A , B , and C .

Braces $\{\}$ are always used when writing sets.

$$A = \{2, 4, 6\}$$

Element

An **element** is an object that belongs to a set.



Symbols for Elements

The symbol $\in$ means “is an element of.”

The symbol $\notin$ means “is not an element of.”

Let $A = \{2, 4, 6\}$.

$$4 \in A \qquad 5 \notin A$$

Roster Form

A set may be written in **roster form**, where all elements are listed inside braces.

$$A = \{0, 1, 2, 3, 4\}$$

Rule Form

A set may also be written in **rule form** (or set-builder form), where a rule describes the elements of the set.

$$A = \{x \mid x \text{ is a whole number less than } 5\}$$

Reading Rule Form

$$A = \{x \mid x \text{ is a whole number less than } 5\}$$

Read as: “ A is the set of all x such that x is a whole number less than 5.”

Same Set, Two Ways

Roster form

$$A = \{0, 1, 2, 3, 4\}$$

Rule form

$$A = \{x \mid x \text{ is a whole number less than } 5\}$$

Memory Tip

Roster = a **list**, like a class roster of names.

Rule = a **description** that tells which elements belong.

Empty Set

The **empty set** has no elements and is written as

$\emptyset$ or $\{\}$

Careful!

$$\{0\}$$

is **not** the empty set, because it has one element: 0.

Examples

Example 1

Write the set of whole numbers greater than 2 but less than 7 in roster form and rule form.

Roster form

$$\{3, 4, 5, 6\}$$

Rule form

$$\{x \mid x \text{ is a whole number greater than 2 but less than 7}\}$$

Example 2

Write the set of multiples of 3 from 3 to 15 in roster form and rule form.

Roster form

$$\{3, 6, 9, 12, 15\}$$

Rule form

$$\{x \mid x \text{ is a multiple of 3 from 3 to 15}\}$$

Example 3

Write the set of months of the year that begin with the letter “M” in roster form and rule form.

Roster form

$\{\text{March, May}\}$

Rule form

$\{x \mid x \text{ is a month that begins with “M”}\}$

Example 4

Let $C = \{r, o, s, e\}$. Is each statement true or false?

(a) $o \in C$ (b) $t \in C$ (c) $e \notin C$

(a) True (b) False (c) False

Example 5

Let $E = \{2, 4, 6, 8\}$. Complete each statement using $\in$ or $\notin$.

$$8 \text{ ______ } E \qquad 3 \text{ ______ } E$$

$$8 \in E \qquad 3 \notin E$$

Example 6

Write the set of days of the week that begin with the letter “B.”

No day of the week begins with “B,” so the set is the empty set.

$\emptyset$

Practice Exercises

Practice 1

Write the set of even whole numbers less than 12 in **roster form** and **rule form**.

Practice 2

Write the set of multiples of 4 from 4 to 24 in **roster** and **rule form**.

Practice 3

Let $B = \{m, a, t, h\}$. Determine whether each statement is true or false.

$$(a) \ a \in B \qquad (b) \ x \in B$$

$$(c) \ m \notin B \qquad (d) \ h \in B$$

Practice 4

Write the set of odd numbers greater than 5 but less than 16 in both roster and rule form.

Practice 5

Write the set of months of the year that begin with the letter “J” in both roster and rule form.

Practice 6

Let $D = \{1, 3, 5, 7, 9\}$. Complete each statement using $\in$ or $\notin$.

(a) $7 \text{ ______ } D$

(b) $4 \text{ ______ } D$

(c) $1 \text{ ______ } D$

(d) $8 \text{ ______ } D$

Activities

Activity 1

Write the set of odd whole numbers less than 14 in **roster form** and **rule form**.

Activity 2

Write the set of multiples of 5 from 5 to 30 in **roster** and **rule form**.

Activity 3

Let $F = \{c, l, a, s\}$. Determine whether each statement is true or false.

$$(a) \ l \in F \qquad (b) \ b \in F$$

$$(c) \ s \notin F \qquad (d) \ a \in F$$

Activity 4

Write the set of even numbers greater than 8 but less than 21 in both roster and rule form.

Activity 5

Write the set of days of the week that begin with the letter “T” in both roster and rule form.

Activity 6

Let $H = \{12, 14, 16, 18, 20\}$. Complete each statement using $\in$ or $\notin$.

(a) 14 ____ H (b) 17 ____ H

(c) 20 ____ H (d) 9 ____ H

Thank you!